SUPERSEDED BY
2000-F13-PID-PR-10011_MS

NOTES:

1. VENDOR SUPPLIED VALVES FIELD MOUNTED AS CLOSE AS POSSIBLE TO BEARING SEALS.
2. CONNECTIONS INDICATED ARE VENDOR TERMINATIONS. MATING FITTINGS TO BE SUPPLIED BY THE PIPING CONTRACTOR U.N.O.
3. DIVIDER TO BE LOCATED IN CLOSE PROXIMITY TO TRUNNION BEARING HOUSE.
4. CALCULATE THE DIFFERENCE BETWEEN THE HIGHEST AND LOWEST TEMPERATURE OF THE THREE SIGNALS.
5. REFER TO VENDOR P&ID: 2000-ME001.3-D07-50002 FOR DETAILS.
6. FOR VALVE BANK DETAILS REFER TO 2000-F13-PID-PR-10013.

COOLING WATER HEADER 2000-F13-PID-PR-10009 COOLING WATER
13-0114-COW-MSG1-100-TI
FE 13413 / FIT 13413 / FI 13413
100VBFD1L2 F13-10322
100VBFD1L2 F13-10323
25VB03S1 F13-10326
DETAIL 1 TCV 13416 FO
100VBFD1L2 F13-10325
100VCD1C1 F13-10328
25VB03S1 F13-10327
TO GRADE
100VBFD1L2 F13-10324
13-1020-COW-MSG1-100-TI
100VBFD1L2 F13-10329
13-2072-COW-MSG1-100
100VBFD1L2 F13-10371
13-0111-COW-MSG1-100
COOLING WATER RETURN HEADER 2000-F13-PID-PR-10009 COOLING WATER RETURN

COOLING WATER HEADER 2000-F13-PID-PR-10009 COOLING WATER
13-0146-COW-MSG1-100-TI
100VBFD1L2 F13-10319
25VB03S1 F13-10320
TO GRADE
100VCD1C1 F13-10321 100x80
13-0111-COW-MSG1-100
100x80
13-1364-LO-SS3-25
13-1365-LO-SS3-25
25VB03S1 F13-10330
TO GRADE

F13-FN-453 BALL MILL LUBE ROOM FAN (INDUCTION)
F13-FN-454 BALL MILL LUBE ROOM FAN (EXTRACTION)
FIXED FLOATING PINION BEARING LUBRICATION
FI 13911 / FI 13910 / FI 13909 / FI 13908
LUBE OIL RESERVOIR
TI 13866 / LI 13874 / TI 13873 / LI 13872

F13-LU-423 BALL MILL MAIN BEARINGS LUBRICATION UNIT

PINION GEAR FLANK TEMPERATURE (RIGHT SIDE)
TI 13955 / VI 13972 / TI 13957 / VI 13974 / TDI 13763 / TI 13763 / TI 13764 / TI 13765 / TI 13766
SP HS-053 / SP HS-049 4 OFF
13-1378-LO-SS1-80
13-2354-LO-SS1-80

LUBE OIL DISTRIBUTION
ZI 13880 / ZSC 13880 / ZSO 13880 / PDI 13879 / TIC 13880 / FI 13881

HP LUBE PUMPS
SAL 13865 / SAL 13864 / SAL 13863 / FI 13891
THRUST RAIL LUBE
FI 13892

BALL MILL MAIN BEARINGS LUBRICATION CONTROL PANEL
HS 13875B / HS 13875C / HS 13877B / HS 13877C / HS 13903B / HS 13903C / HS 13905B / HS 13905C / HS 13885B / HS 13885C / HS 13887B
HS 13887C / HS 13889B / HS 13889C / HS 13925A / HS 13925B / HS 13925C / HS 13968A / HS 13968B / HS 13968C / HS 13914 / HS 13915
HS 13913 / HS 13953 / XI 13912 / XI 13937 / XI 13940 / XI 13943 / XI 13946 / XI 13949 / XI 13952 / XI 13976 / XI 13916
XI 13917 / XI 13918 / XI 13920 / XI 13921 / XI 13922 / XI 13923 / XI 13919 / XI 13924 / XI 13926 / XI 13927 / XI 13928
XI 13929 / XI 13930 / XI 13931 / XI 13932 / XI 13933 / XI 13934 / XI 13935 / XI 13936 / XI 13941 / XI 13942 / XI 13951
XI 13948 / XI 13950 / XI 13947

FEED 13-1354-LO-SS9-38 / 13-1355-LO-SS9-38 / 13-1356-LO-SS9-38 / 13-1357-LO-SS9-38 — VALVE BANK TYPE 1
THRUST 13-1358-LO-SS9-38 / 13-1359-LO-SS9-38 / 13-1360-LO-SS9-38 / 13-1361-LO-SS9-38 — VALVE BANK TYPE 2
DELIVERY 13-1350-LO-SS9-38 / 13-1351-LO-SS9-38 / 13-1352-LO-SS9-38 / 13-1353-LO-SS9-38 — VALVE BANK TYPE 1
NOTE 6
SP HS-047 4 OFF
SP HS-048 4 OFF
SP HS-047 4 OFF
SP SS-033
SP SS-034

F13-MB-401 BALL MILL NOTE 5
DELIVERY END
FEED END

OCM UNIT
START XC 13201A / STOP XC 13201B / ALARM XC 13201
RETURN DN200
13-1362-LO-SS1-200
RETURN
BALL MILL SEAL GREASE SYSTEM 2000-F13-PID-PR-10013 GREASE
MILLS RING GEAR SPRAY SYSTEM 2000-F13-PID-PR-10013 AIR
MILLS RING GEAR SPRAY SYSTEM 2000-F13-PID-PR-10013 GREASE
13-1077-LO-SS8-16-ET
13-1072-LO-SS8-16-ET
13-1496-IA-MSG3-15
13-1440-LO-SS8-20-ET
13-1249-LO-SS8-16-ET
13-1250-LO-SS8-16-ET
ET
KI 13494 / KI 13493 / KI 13491 / KI 13492
SP HS-019
20x12
13-1363-LO-SS1-200

PINION GEAR FLANK TEMPERATURE (LEFT SIDE)
TI 13954 / TI 13956 / VI 13973 / VI 13971 / TDI 13786 / TI 13783 / TI 13784 / TI 13785 / TI 13786
SP HS-053 2 OFF
SP HS-049 4 OFF
13-1379-LO-SS1-80
13-2355-LO-SS1-80

GIRTH GEAR GUARD
13-0136-FD-MS1-100
WASTE SKIP

MOTOR BEARING RTD (INDE) VI 13767 / MOTOR BEARING RTD (DE) VI 13770
TIC 13414 / TIT 13414 / TE 13414
DN100
80x65
F13-MB-401A BALL MILL MOTOR
13-1368-LO-SS8-20
13-1369-LO-SS8-20
13-1376-LO-SS1-80
13-0138-COW-MSG1-100
25VB03S1 F13-10332
100VBFD1L2 F13-10331
TO GRADE
COOLING WATER RETURN HEADER 2000-F13-PID-PR-10009 COOLING WATER RETURN

FEED TRUNNION BEARING RTDS (RIGHT SIDE)
NOTE 4 TDI 13964 f(x)
TI 13964A OUTBOARD / TI 13964B CENTRE / TI 13964C INBOARD
DISCHARGE TRUNNION BEARING RTDS (RIGHT SIDE)
NOTE 4 TDI 13958 f(x)
TI 13958A OUTBOARD / TI 13958B CENTRE / TI 13958C INBOARD
THRUST BEARING (RIGHT SIDE)
TI 13962A OUTBOARD / TI 13962B INBOARD

FEED TRUNNION BEARING RTDS (LEFT SIDE)
NOTE 4 TDI 13966 f(x)
TI 13966A OUTBOARD / TI 13966B CENTRE / TI 13966C INBOARD
DISCHARGE TRUNNION BEARING RTDS (LEFT SIDE)
NOTE 4 TDI 13960 f(x)
TI 13960A OUTBOARD / TI 13960B CENTRE / TI 13960C INBOARD
THRUST BEARING (LEFT SIDE)
TI 13967A OUTBOARD / TI 13967B INBOARD

MOTOR BEARING RTD (INDE) VI 13789
MOTOR BEARING RTD (DE) VI 13792
TIC 13416 / TIT 13416 / TE 13416
F13-MB-401B BALL MILL MOTOR 2
100VCD1C1 F13-10318
DN100
13-0139-COW-MSG1-100
13-1370-LO-SS8-20
13-1371-LO-SS8-20
80x65
13-1374-LO-SS1-80
13-1375-LO-SS1-80
13-1377-LO-SS1-80
25VB03S1 F13-10334
100VBFD1L2 F13-10333
TO GRADE
COOLING WATER RETURN HEADER 2000-F13-PID-PR-10009 COOLING WATER RETURN

LUBE OIL RESERVOIR
LAL 13819 / LALL 13819 / TSH 13820 / TSL 13820 / PDAH 13816 / PDAHH 13816 / PDAH 13817 / PDAHH 13817 / TI 13821 / LI 13822

F13-LU-421 BALL MILL MOTORS LUBRICATION UNIT
80x40
MOTOR 1 DE/NDE LUBE OIL SUPPLY
FI 13827 / FI 13828 / PAH 13823 / PI 13823 / PAH 13824 / PI 13824
MOTOR 2 DE/NDE LUBE OIL SUPPLY
FI 13829 / FI 13830 / PAH 13825 / PI 13825 / PAH 13826 / PI 13826

13-0115-COW-MSG1-100-TI
COOLING WATER HEADER 2000-F13-PID-PR-10009 COOLING WATER
FI 13415 / FIT 13415 / FE 13415
100VBFD1L2 F13-10312
100VBFD1L2 F13-10313
25VB03S1 F13-10316
TO GRADE
DETAIL 1 TCV 13416 FO
100VBFD1L2 F13-10315
25VB03S1 F13-10317
TO GRADE
13-0310-COW-MSG1-100-TI
100VBFD1L2 F13-10314
13-1363-LO-SS1-200
13-2073-COW-MSG1-100
100VBFD1L2 F13-10312